

ANMOL TANWAR

GIStructE | Graduate Structural Engineer

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PROFESSIONAL SUMMARY

Ambitious and creative Graduate Structural Engineer (GIStructE) with 2+ years of UK experience delivering technically demanding structural projects across healthcare, commercial, and residential sectors. Proficient in structural analysis and design of concrete, steel, timber, and masonry structures to BS-EN Eurocodes, British Standards, and Indian Standard with hands-on experience in FEM tools including Tekla, STAAD Pro, ETABS, A self-motivated problem-solver committed to excellence, precision, and continuous professional growth, actively progressing toward MStructE chartership. I passed the IStructE Chartership Exam in April 2026.

CORE COMPETENCIES

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| • Structural Analysis & Design (BS-EN Eurocodes / British Standards/ Indian Standard Codes) | • FEM Tools: STAAD Pro / ETABS / Midas Gen / Tekla Structural Designer | • Steel, Concrete, Timber & Masonry Design |
| • Documentation: RIBA Stage 1–5 Structural Reports | • AutoCAD / BIM Workflows Coordination | • Computational Design: Python / MATLAB |
| • Interdisciplinary Team Collaboration | • Shop Drawing Review & Design Coordination | • Problem Solving & Precision Detailing |

WORK EXPERIENCE

C.C.S Consulting – Graduate Structural Engineer (GIStructE) (January 2024 – January 2026)

Manchester, United Kingdom

Delivered structural design and technical engineering across major NHS hospitals refurbishment and new build schemes, industrial and residential projects in Greater Manchester, working within live operational facilities throughout all construction phases. My key responsibilities involved:

- Developed structural designs from RIBA Stage 1 (Concept) to RIBA Stage 5 (Construction) for concrete, steel, timber and masonry building. Design using hand calculation and Tekla Tedds for various structural elements, modelling of structures in FEM software (Tekla Structural Designer).
- Embodied carbon calculation using IStructE tools.
- Coordinating structural design across interdisciplinary teams including architectural, MEP, liaising with contractors directly and the wider project team
- Carried out embodied carbon calculations, produced engineering reports, and attended design team meetings and site visits across multiple project stages simultaneously
- Preparing design briefs for subcontractor elements, site investigation requirements, and reviewed the fabrication drawings.
- Conducted structural surveys, reviewed site investigations report

Additionally, developed automation tools using coding to streamline engineering workflows and improve efficiency within the organisation, and delivered internal CPD sessions to share technical knowledge and promote continuous professional development among colleagues.

Key Projects

Wythenshawe Maternity Refurbishment, UK (Value: Approx. 15 million GBP)

- Structural engineer on health care scheme with NHS to facilitate refurbishment of wards & operating theatres, extension to existing building, temporary modular buildings, civil design, roof replacement and demolition of existing RAAC roof structure.
- Client-side checking role and preparation of design intent for the sub-contractor design such as temporary over roofing scaffolding, crash deck, review fabrication drawings, connection design drawings/ calculations etc.
- Contributed to structural assessment and risk mitigation of Reinforced Autoclaved Aerated Concrete (RAAC) structures across multiple hospitals in North West England.
- Delivered structural analysis and retrofit strategies aimed at collapse prevention and ensuring operational safety within critical healthcare facilities.
- Supported engineering decision-making on structural strengthening, monitoring strategies, and long-term remediation options for ageing infrastructure.

Wythenshawe Endoscopy, UK (Value:-)

- Engineered structural modifications to existing flat & two-way slabs while maintaining full operational continuity on the live 1st floor as a part of refurbishment scheme.
- Designed roof-level gantry frame (including connections) to support new AHU, plus site investigations and targeted strengthening works.

Manchester Royal Infirmary, UK (Value: -)

- Concept design to Stage 3 design for Project Red – Supply which required steel towers, gantry frames, supports at the roof structure, modification to the existing structure, to facilitate the service runs to supply project Red.
- Specified temporary works and investigation scope to enable complex service routing with minimal hospital disruption.
- Liaising with contractor and design team. Review site investigations

North Manchester General Hospital, UK (Value: -)

- As a part of the scheme, involved in the design team meeting with client/contractor to prepare the proposals of concrete repairs for the existing concrete gantry repair works, investigation requirement, loading appraisals of the building built in 90's.
- Client-side checking role for the preparation of design intent briefs for various scaffolding proposals (sub-contractor design) around the existing 3 storey + basement brick masonry building to accommodate the refurbishment works in the 3 floor wards, loading appraisal for the existing timber floor to specify the limit loading to the existing floor from crash decks.

- Structural surveys to identify the existing structural build-ups at required locations.
- RAAC surveys, short term remedial design, inspections and roof replacements

C.C.S Consulting – Intern Structural Engineer (*August 2023 – December 2023*)

Manchester, United Kingdom

- Conducted structural surveys of existing and listed buildings to review design proposals, performed hand calculations for concrete, steel, timber, and masonry structures to BS-EN Eurocodes.
- Prepared drawings across RIBA Stages 1–5 using AutoCAD, assisted senior engineers with technical report writing and design calculations, demonstrating precision and attention to detail.
- Prepared stability calculations and technical reports for temporary demountable structures.

P.C. Ragtah and Associates – Intern Structural Engineer (*January 2022 – July 2022*)

India

- Modelled and analysed structures using STAAD Pro and ETABS (FEM tools), assisting with structural design of RC foundations, retaining walls, beams, columns, slabs, and staircases under senior engineer guidance.
- Assisted in preparation of design and construction documents for Stage 4 and Stage 5 deliverables, drafting project drawings and coordinating details and dimensions across disciplines.

Municipal Corporation of Delhi – Quality Check Trainee (*June 2018 – July 2018*)

India

- Conducted comprehensive material testing on an active 100-bed construction site, verifying compliance of concrete and other structural materials against Indian Standard (IS) codes to maintain quality assurance throughout the project.

EDUCATION

MSc Structural Engineering

2022 – 2023

The University of Manchester, United Kingdom | GPA: 2.59/4.00

Concentration: Static and dynamic behaviour of steel and concrete structures, earthquake and fire engineering.

- Dynamic Behaviour Study: Recorded, post-processed, and interpreted acceleration data on Merchants Bridge, Manchester using NI myRIO and LabVIEW/MATLAB demonstrating computational analysis skills. (Score: 77.5/100)
- Concrete Centre Design Competition: Led an interdisciplinary team of 6 in designing a 6-storey + basement office building using hybrid in-situ and precast concrete, balancing structural innovation with environmental sustainability. Final report: 84.2/100.

B.Tech Civil Engineering

2019 – 2022

Shree Guru Gobind Singh Tricentenary University, India | GPA: 3.15/4.00

Concentration: Structural analysis and design of concrete and steel structures, sustainable development.

- Major project: Industrial training (6 months) and design of a G+4 residential building.

TECHNICAL SKILLS & SOFTWARE

Structural Software	STAAD Pro, ETABS, Tekla Structural Designer, Midas Gen, Ansys, Robot (exposure)
BIM & Drawing	Revit Structure (working knowledge), AutoCAD (proficient), BIM360/ACC (working knowledge)
Design Codes	BS-EN Eurocodes (EN 1990–1999), British Standards, Indian Standards
Programming	Python (working knowledge), MATLAB (proficient), Dynamo (awareness)
Other Tools	Tedds, MS Office Suite (Excel, Word, PowerPoint)
Sustainability	LEED Green Associate (basic knowledge), BREEAM awareness

TRAINING & CERTIFICATIONS

- Structural Behaviour Exam (IStructE)
- Tekla Structural Designer Fundamentals (*Jan 2025, March 2024*)
- Footfall Assessment in Tekla Structural Designer (*Dec 2023*)
- SGT University – Industrial Trainee
 - Experimented with and analysed the behaviour of structural members under various loading conditions using Abaqus finite element software.

VOLUNTEERING EXPERIENCE**CROSS UK Early Career Volunteer** (*April 2026- Present*)

- Attending expert panel quarterly meetings
- Contribution to the production of CROSS Safety Reports, presentation to the expert panels and preparation of communication materials

IStructE – Chair, Northwest Young Members Group (*January 2026 – Present*)

- Currently leading the Northwest Young Members Group, setting strategic direction and fostering professional development among emerging structural engineers.
- Chairing monthly committee meetings, setting agendas, facilitating structured discussions, and driving productive outcomes.
- Planning and coordinating professional events and site visits, managing logistics, scheduling, and regional group communications.
- Maintaining active coordination across the team to ensure members remain engaged and aligned with group objectives.

IStructE – Vice Chair, Northwest Young Members Group (*January 2025 – December 2025*)

- Collaborated with Prof. Jonathan Hayes at the University of Salford to contribute to the development of a structural behaviour course for undergraduate and postgraduate engineering students as a volunteer
- Supported the leadership of the young members group, assisting in the organisation and delivery of professional events and site visits.
- Coordinated actively with team members to ensure smooth communication and effective delivery of group activities.

University of Manchester

- Led a team of 6 colleagues in completing the Concrete Centre design competition project for a 6-storey office building.
- Served as MSc Structural Engineering Student Representative, liaising between students and academic staff.

SGT University, India

- Led a student team in organising and managing the ROBO WAR event at the university's annual Synergy event.

AWARDS AND ACHIEVEMENTS

- Best structural design award during MSc Structural Engineering at The University of Manchester
- Delivered a technical presentation at the Institution of Structural Engineers Young Members Presentation event (November 2024). Recognised for the quality of the technical paper and invited to attend the IStructE Hot Pot Supper, receiving a prize of recognition from Mohamed Al-Dah, President of the Institution of Structural Engineers.